



Credit Card

FRAUD DETECTION



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AGENDA



- 01. Introduction
- 02. Background
- 03. Related Work
- 04. Exploratory Data Analysis
- 05. Experiments
- 06. Results
- 07. Conclusions

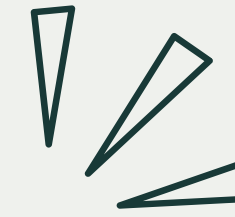




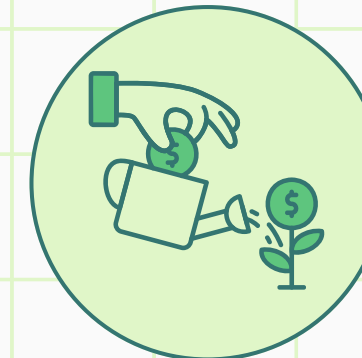
INTRODUCTION

- Credit card fraud is a growing issue in digital payments
- Fraud is rare, with $<1\%$, but causes real financial loss
- Accuracy alone is not effective since it can fail to detect some fraud
- Requires handling class imbalance and better metrics



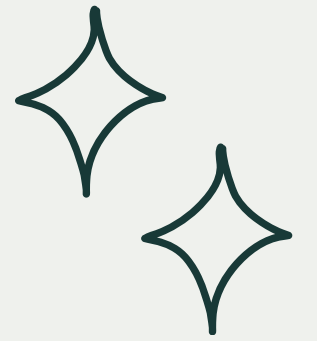


BACKGROUND



- Card-not-present (CNP) fraud is the most common form, stolen card used online
- False negative cost: the full transaction amount plus chargeback fees
- False positive cost: blocked legitimate customer, friction, potential churn
- These two error types have very different real-world consequences
- This asymmetry is why we use cost-sensitive learning

RELATED WORK



- Chawla et al. (2002): introduced SMOTE for class imbalance
- Dal Pozzolo et al. (2015): gradient boosting outperforms LR on fraud data
- Bhattacharyya et al. (2011): Random Forest effective with feature interpretability
- Chen & Guestrin (2016): XGBoost introduced, dominant on tabular data
- Japkowicz & Stephen (2002): studied over- vs undersampling strategies

Over-Sampling (SMOTE)

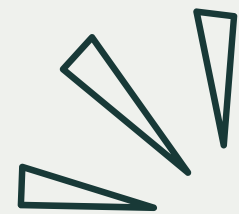


Under-Sampling



Logistic Regression

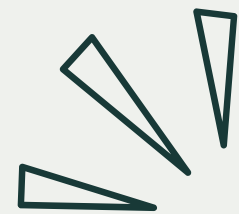




EXPLORATORY DATA ANALYSIS

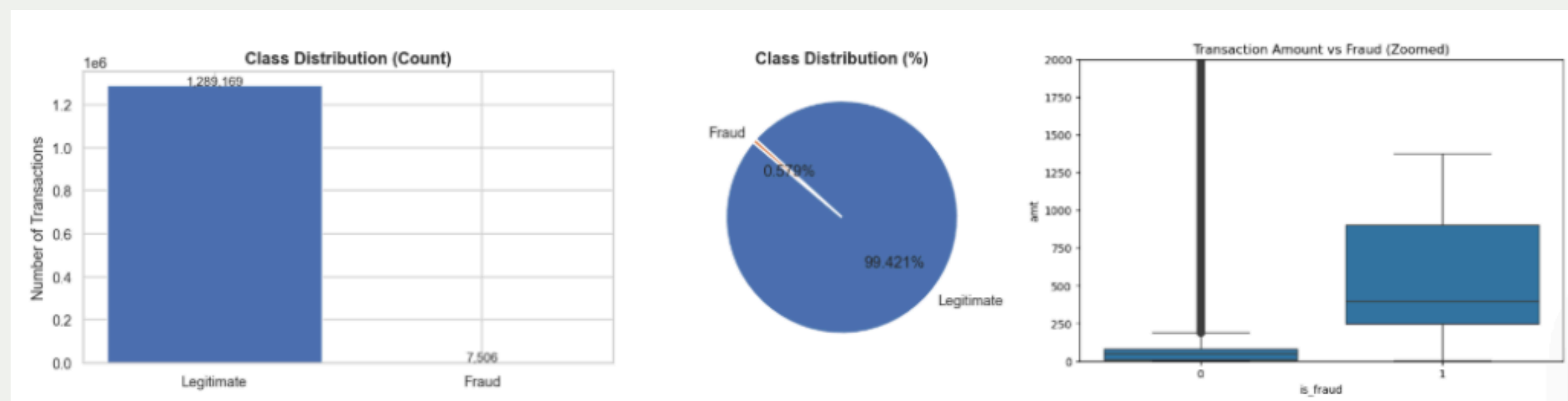
- Kaggle Credit Card Fraud Dataset
- Time Period: Jan. 2019 - Dec. 2020
- ~ 1.8 million transactions
- 1,000 customers, multiple merchants
- Key features:
 - Age, Gender, Job
 - Time-Based: Hour, Day of the Week, and Month
 - Transaction Amount (amt)
 - Category (category_clean)
 - Fraud Label (is_fraud)

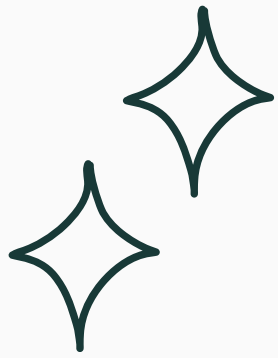




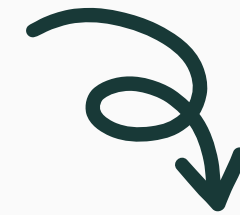
EXPLORATORY DATA ANALYSIS

- Removed irrelevant columns (ex. Name, ID, card number)
- Checked data types and missing values
- No missing value found
- Severe class imbalance:
 - ~99.4% legitimate transaction
 - ~0.6% fraudulent transaction

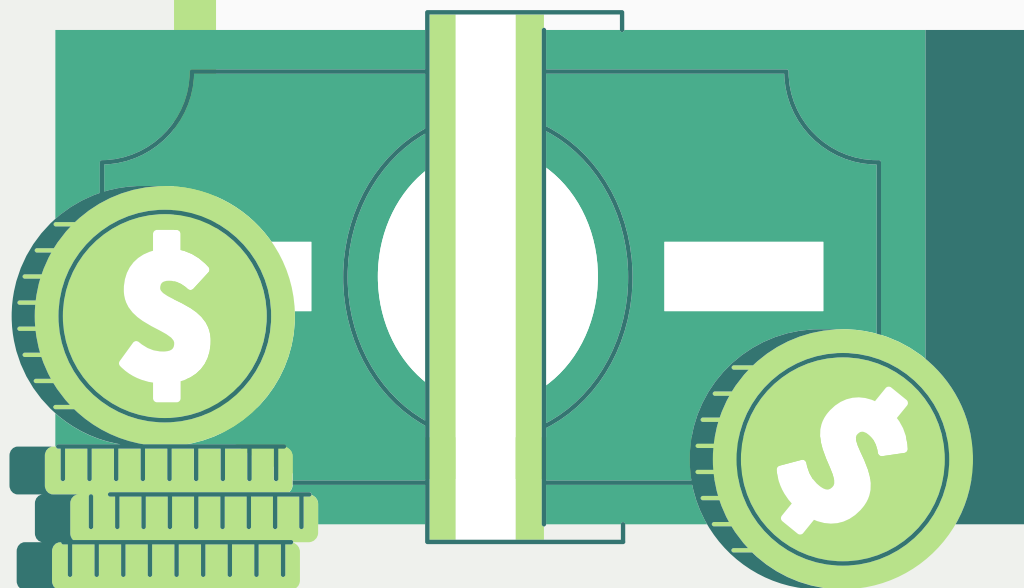
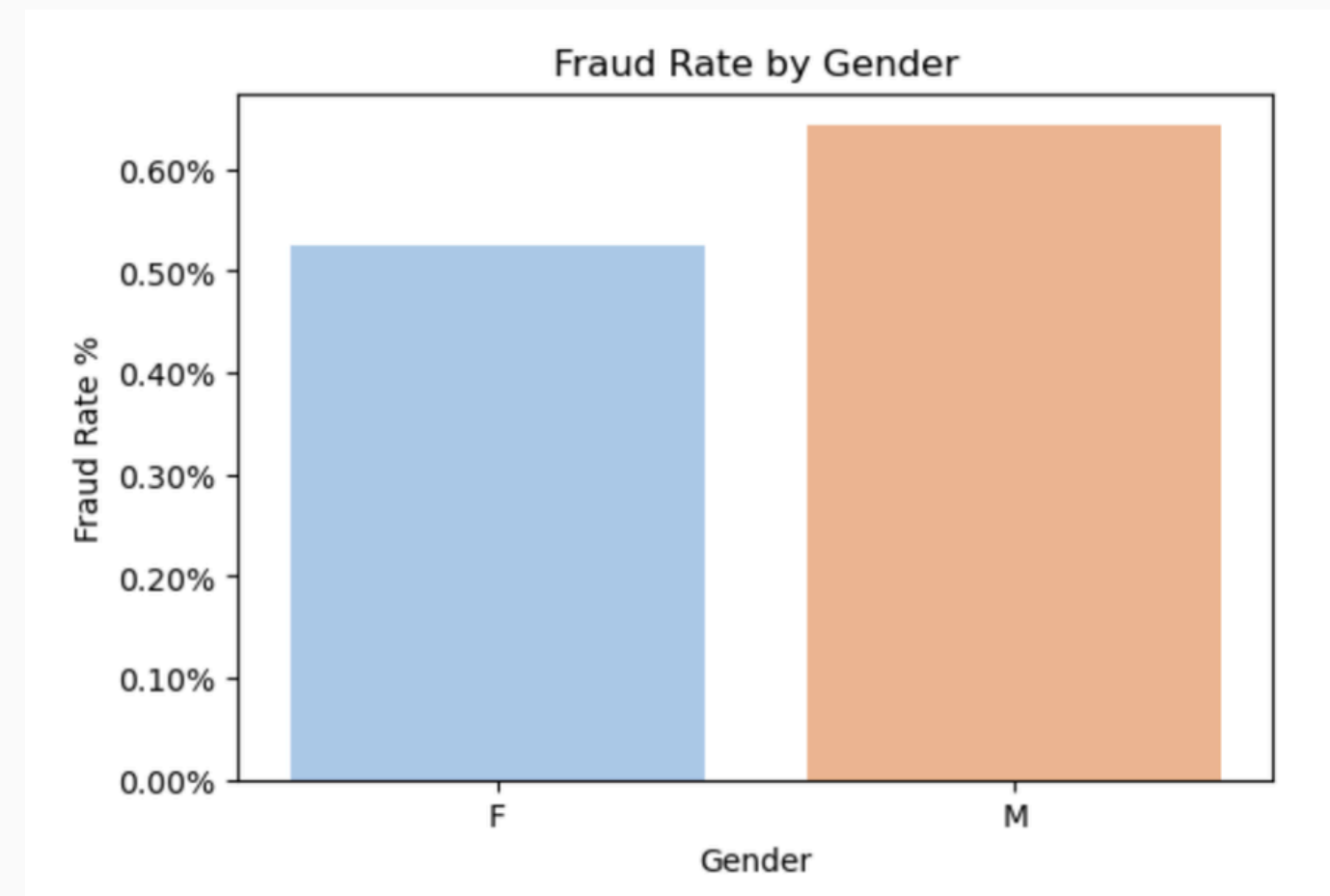


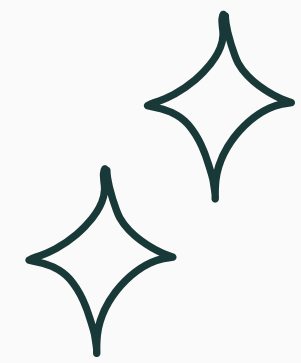


FRAUD RATE BY GENDER

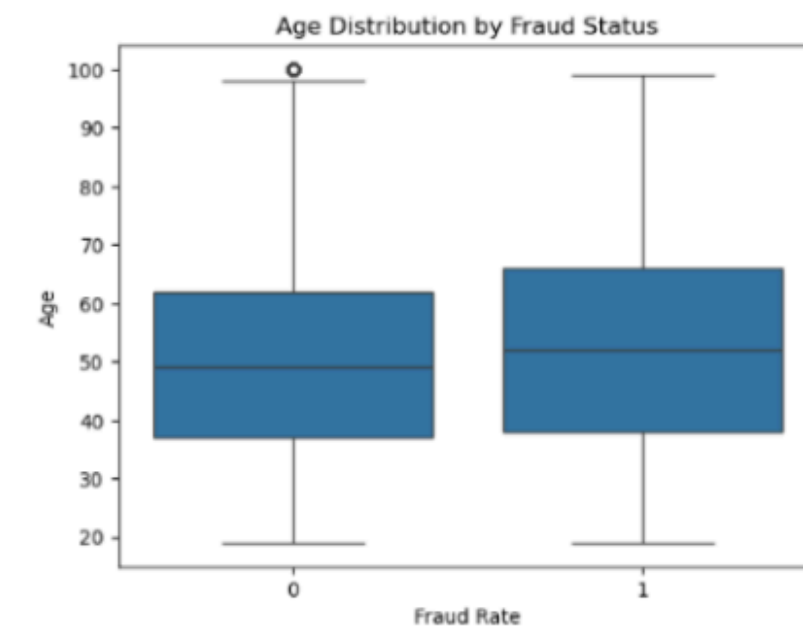
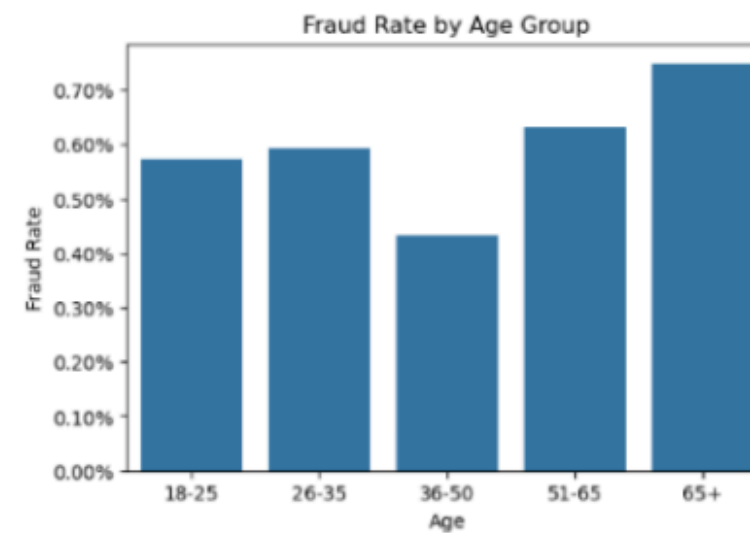
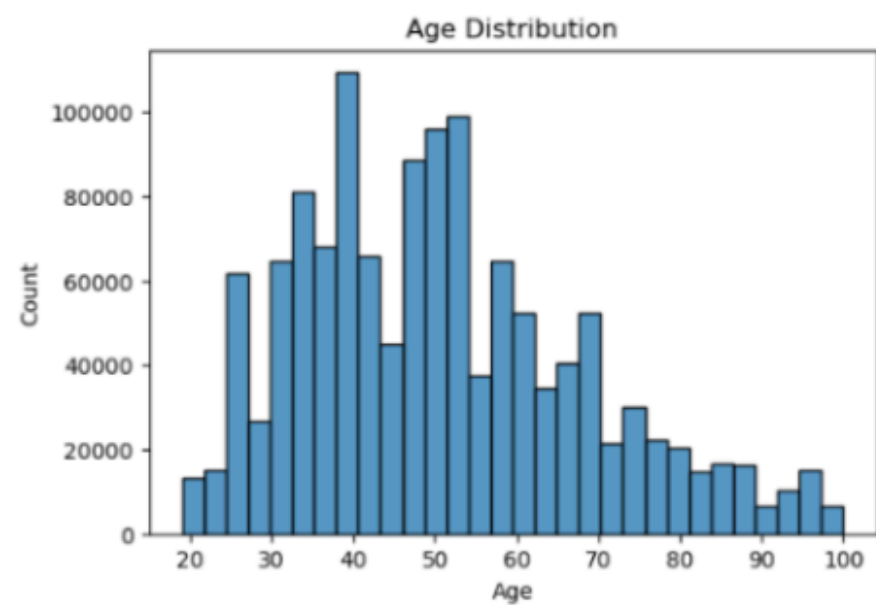


- Female transactions > Male transactions
- Fraud rate slightly higher for males
- Difference is very small (~0.1%)
- Gender is not a strong predictor





AGE DISTRIBUTION & FRAUD PATTERN



- Most users: 30 - 60 years old
- Higher fraud rate in older groups (65+)
- Fraud cases skew slightly older
- Significant overlap → not a strong predictor



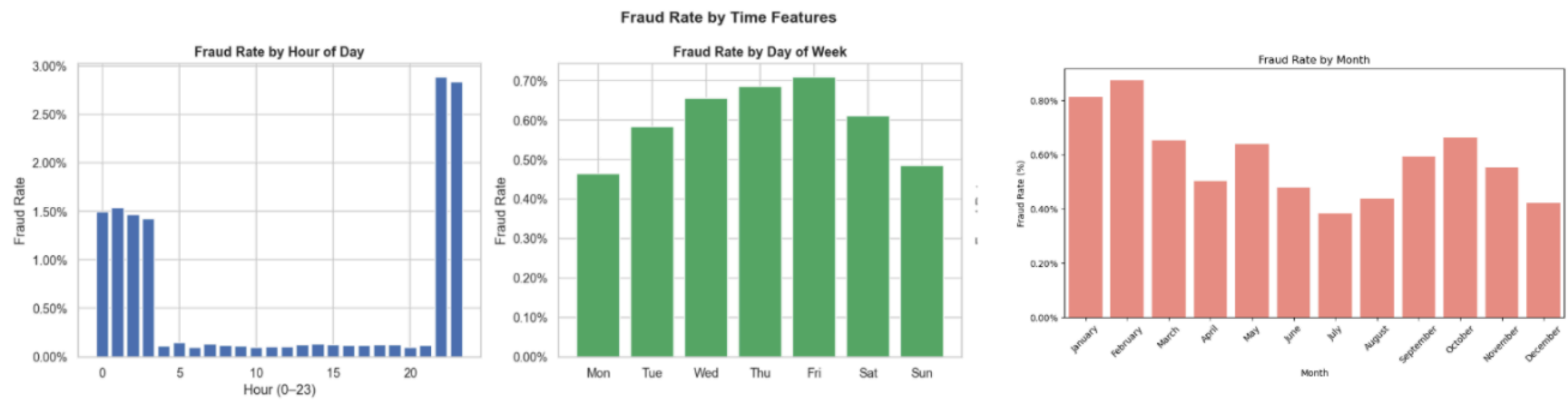
TRANSACTION AMOUNT VS FRAUD



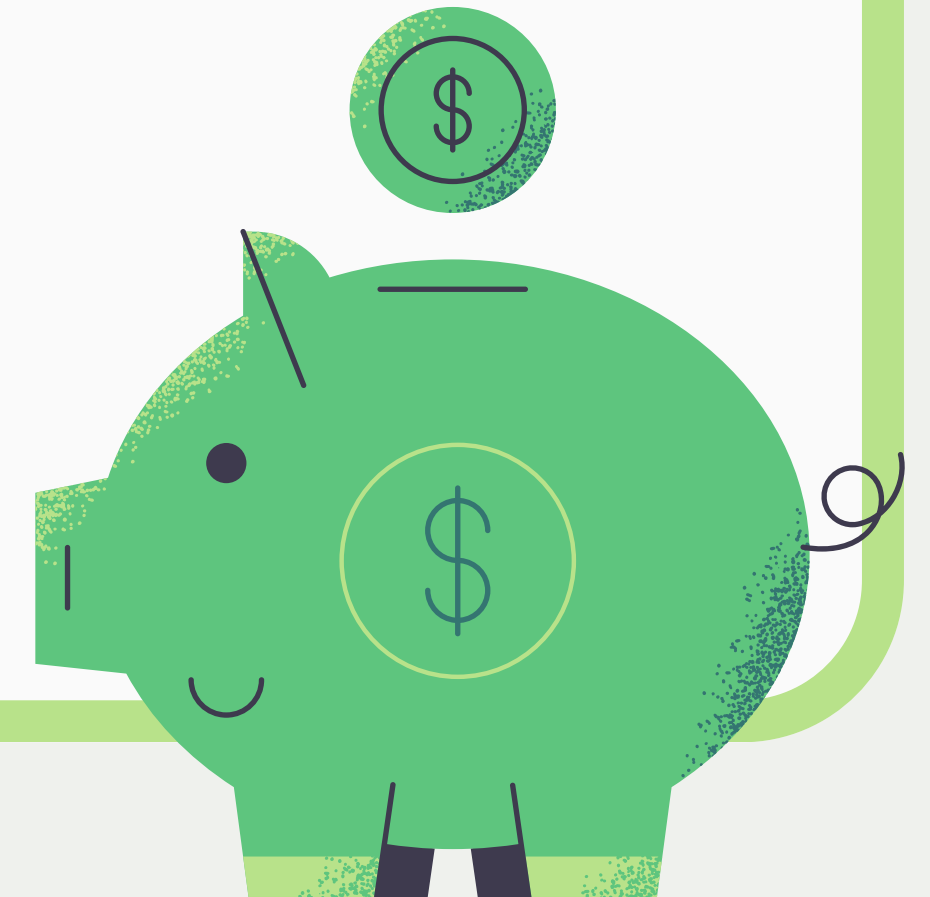
- Fraud transactions tend to have higher amounts
- Normal transactions are mostly low-value
- Some overlap exists between groups
- Amount is a strong predictor of fraud

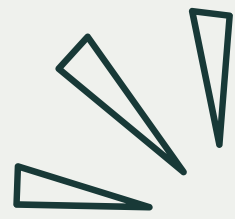


FRAUD PATTERN BY TIME

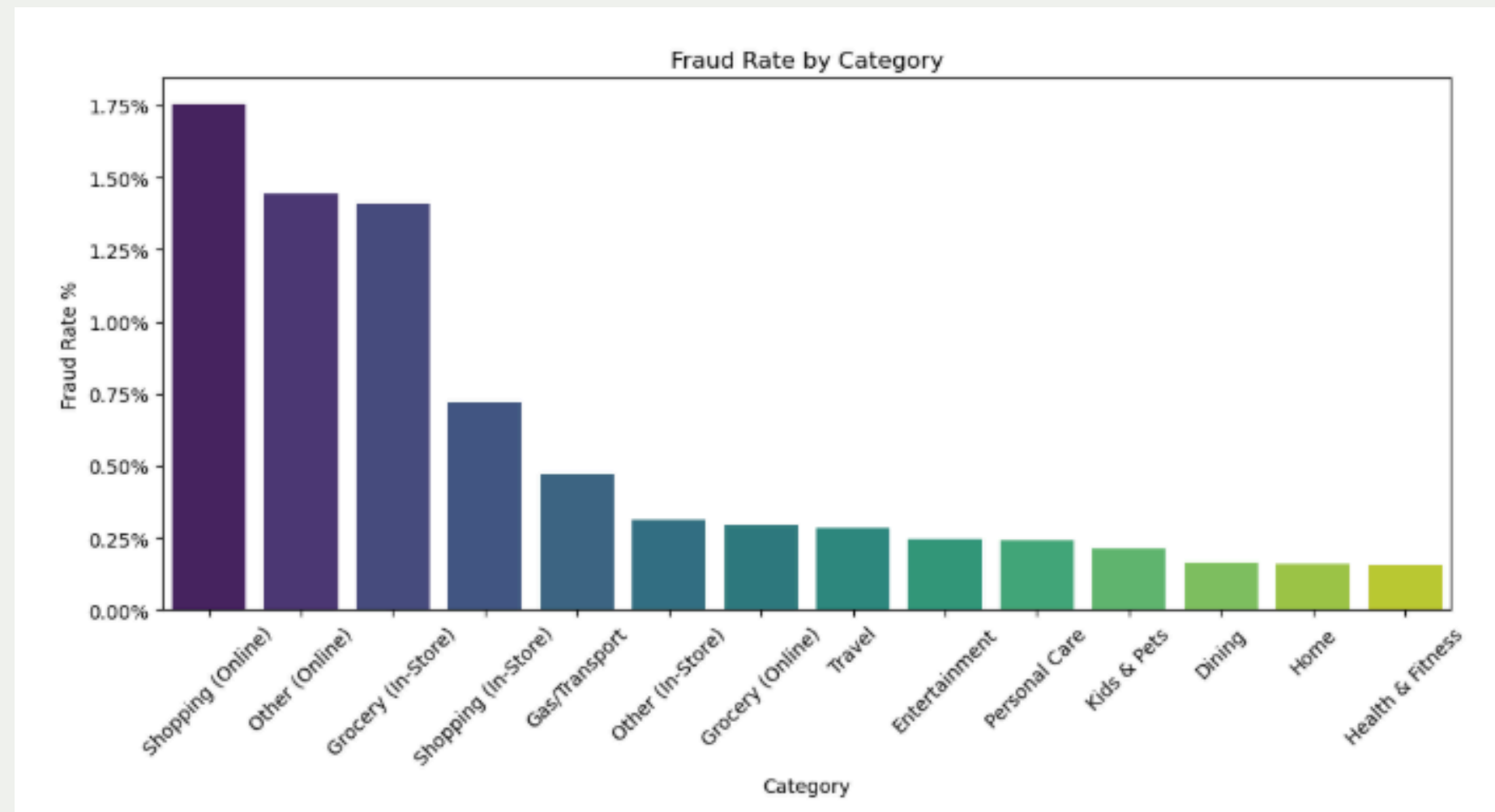


- Higher fraud at late night / early morning (UTC)
- Fraud increases during weekdays
- Slight variation across months
- Time features provide useful signals



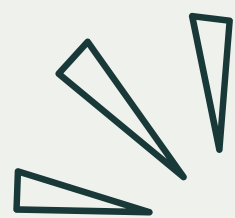


FRAUD RATE BY CATEGORY

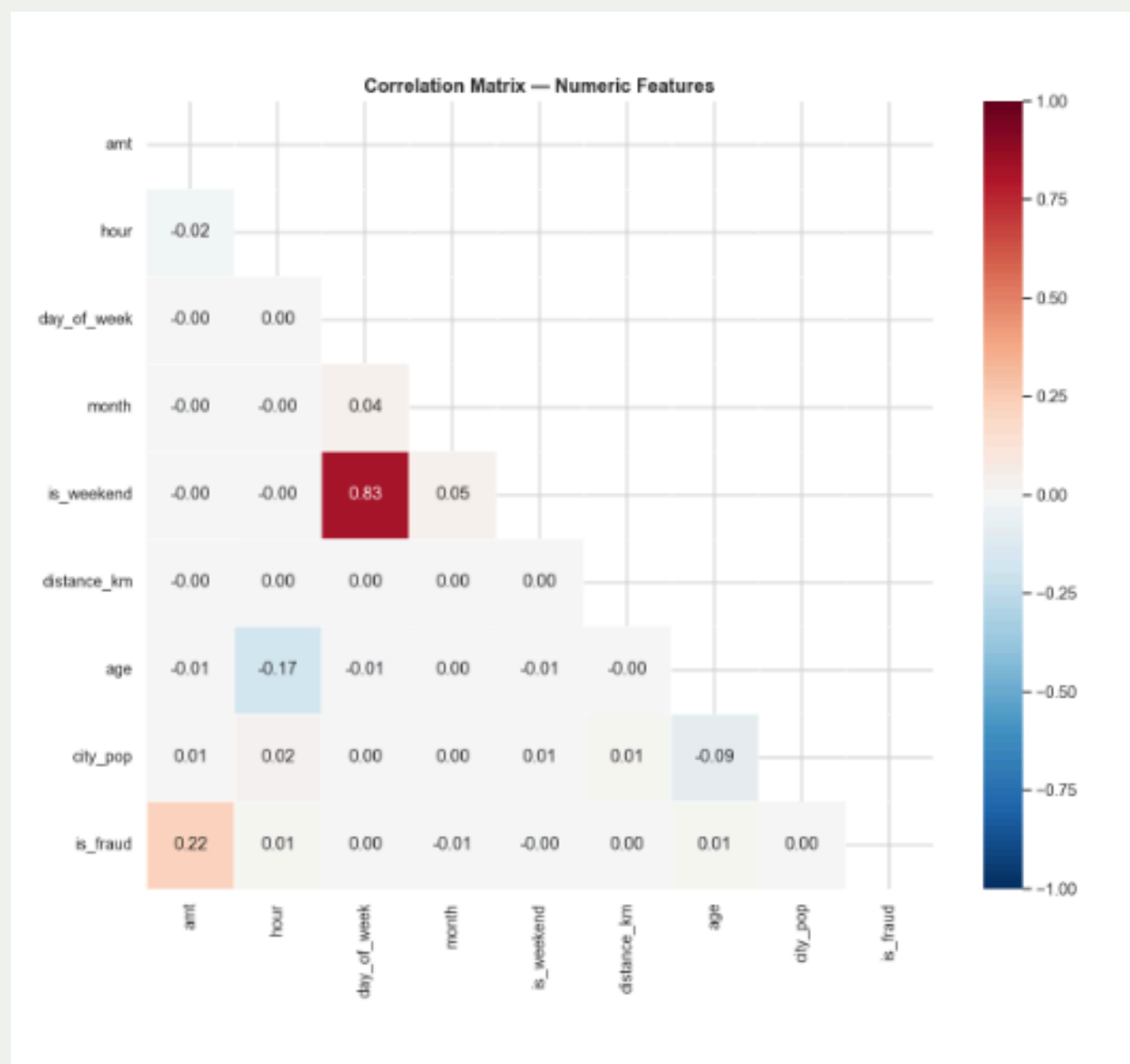


- Online transactions have the highest fraud rates
- Shopping (online) is the most risky category
- In-store transactions have lower fraud rates
- Some categories show minimal fraud activity



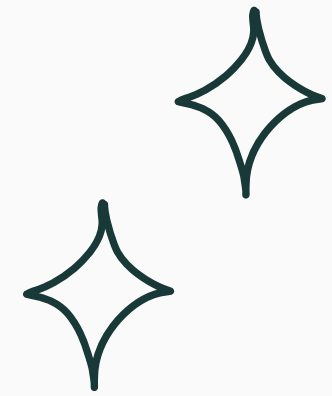


CORRELATION ANALYSIS



- Most features show low correlation with fraud
- No single feature strongly predicts fraud
- Features provide a combined pattern
- Supports the use of advanced models





EXPERIMENT - EVALUATION METRICS



Accuracy is misleading for imbalanced data

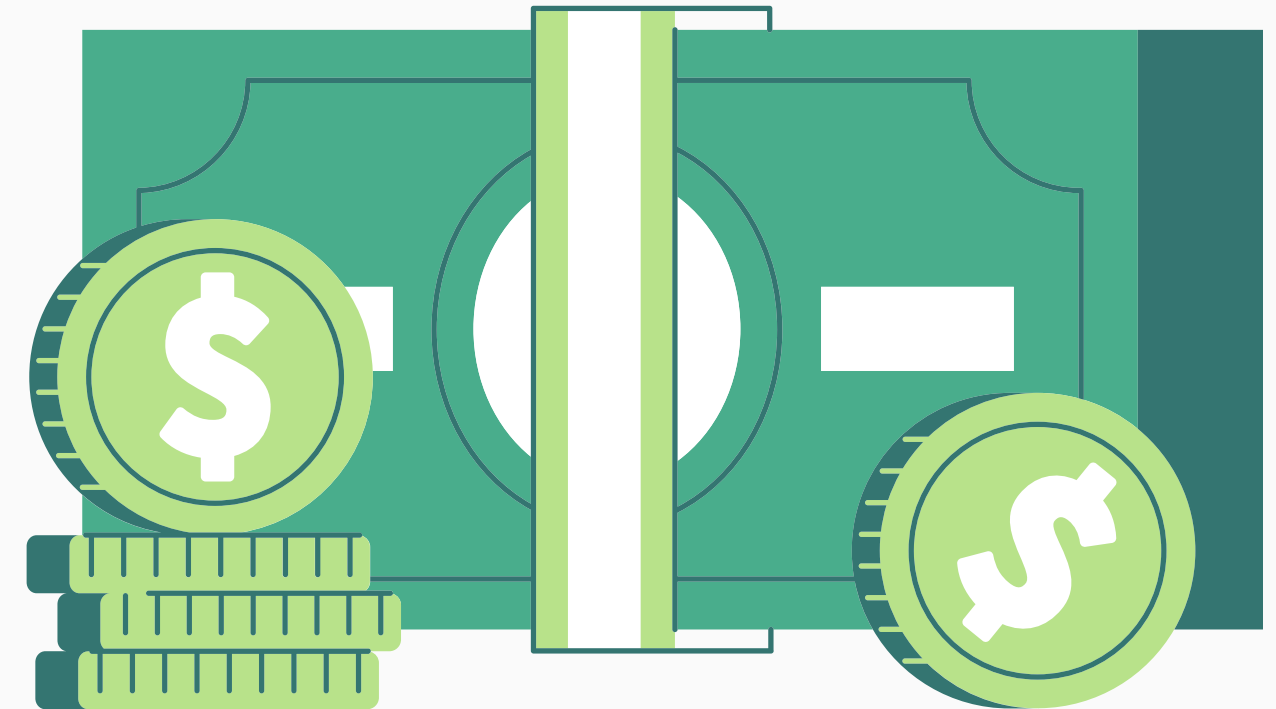
Precision → how correct fraud predictions are

Recall → how much fraud we caught

F1 - score → balance of Precision & Recall

PR - AUC → overall model performance

Confusion Matrix → breakdown of predictions



EXPERIMENT - MODELS IMPLEMENTED

- Logistic Regression (baseline)
 - simple & easy, predicts the probability of fraud
- Random Forest
 - combines multiple decision trees → improves accuracy and reduce overfitting
- XGBoost
 - more advanced model → builds tree sequentially to correct previous errors
- Compare simple vs advanced methods



→ RESULTS - LOGISTIC REGRESSION ✨ ✨

Confusion Matrix:

```
[[222925  34890]
 [   362   1158]]
```

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.86	0.93	257815
1	0.03	0.76	0.06	1520
accuracy			0.86	259335
macro avg	0.52	0.81	0.49	259335
weighted avg	0.99	0.86	0.92	259335

Baseline Model

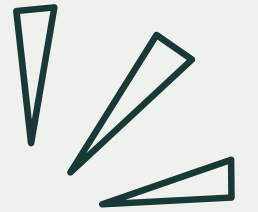
- High accuracy (86%) but misleading
- Recall (Fraud): 0.76 → good detection
- Precision (Fraud): 0.03 → many false positives
- Weak overall performance

UNDER-SAMPLING

- Reduces the majority class (non-fraud)
- Matches minority class (fraud)
- Creates balanced training data
- Helps the model learn fraud patterns
- Trained the model → evaluated on the original imbalanced test



RANDOM FOREST



Random Forest Confusion Matrix: Under-Sampling

```
[[249352  8463]
```

```
[   64  1456]]
```

Random Forest Classification Report: Under-Sampling

	precision	recall	f1-score	support
0	1.00	0.97	0.98	257815
1	0.15	0.96	0.25	1520
accuracy			0.97	259335
macro avg	0.57	0.96	0.62	259335
weighted avg	0.99	0.97	0.98	259335

Feature Importance:

	Feature	Importance
0	amt	0.595954
2	hour	0.213345
13	category_clean_Grocery (In-Store)	0.029980
22	category_clean_Shopping (Online)	0.024999
12	category_clean_Gas/Transport	0.022989
..	...	...
62	state_RI	0.000020
30	state_DC	0.000014
...		
34	state_HI	0.000000
31	state_DE	0.000000

- Precision (fraud): 0.15 → improved, but still low
- Recall (fraud): 0.96 → excellent detection
- Better than Logistic Regression
- “amt” is the most important factor

XGBOOST MODEL

```
XGBoost Confusion Matrix: Under-Sampling
[[252271  5544]
 [   28  1492]]

XGBoost Classification Report: Under-Sampling
              precision    recall  f1-score   support

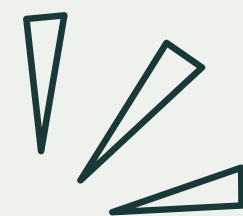
     0       1.00      0.98      0.99    257815
     1       0.21      0.98      0.35      1520

 accuracy          0.98    259335
 macro avg          0.61    259335
 weighted avg       1.00    259335
```

```
Feature Importance:
              Feature  Importance
12  category_clean_Gas/Transport  0.176446
0      amt  0.121996
14  category_clean_Grocery (Online)  0.066862
2      hour  0.053234
11  category_clean_Entertainment  0.045811
..      ...      ...
34      state_HI  0.000000
48      state_MS  0.000000
...      ...      ...
36      state_ID  0.000000
53      state_NH  0.000000
```

- Recall (fraud): 0.98 → excellent
- Precision (fraud): 0.21 → improved
- Best Performance so far
- Key features: category, amount, hour





OVER-SAMPLING (SMOTE)

Synthetic Minority Over-sampling Technique

- Creates synthetic fraud samples
- Balances the dataset without removing data
- Helps the models learn minority patterns
- Risk: may introduce noise



RANDOM FOREST - SMOTE



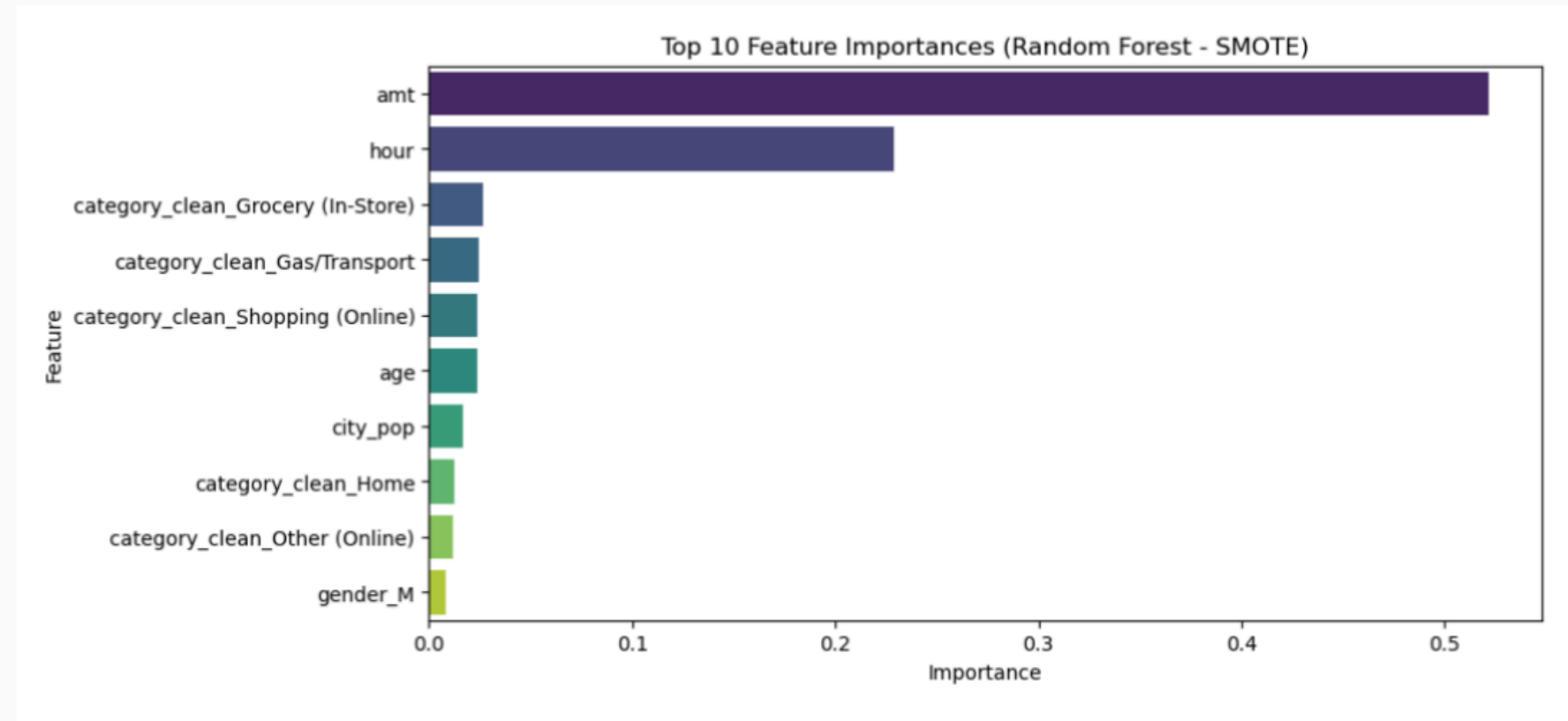
```
Random Forest SMOTE Confusion Matrix:  
[[256307  1508]  
 [   121  1399]]  
  
Random Forest SMOTE Classification Report:  
              precision    recall  f1-score   support  
  
      0         1.00        0.99         1.00    257815  
      1         0.48        0.92         0.63      1520  
  
   accuracy              0.99    259335  
  macro avg              0.74    259335  
weighted avg              1.00    259335
```

- Recall (fraud): 0.92 → strong detection
- Precision (fraud): 0.48 → large improvement
- Better balance than under-sampling
- Fewer false positives overall





RANDOM FOREST - SMOTE



- Transaction amount (amt) is most important
- Time features (hour) are influential
- Transaction category also matters
- Demographics have a lower impact



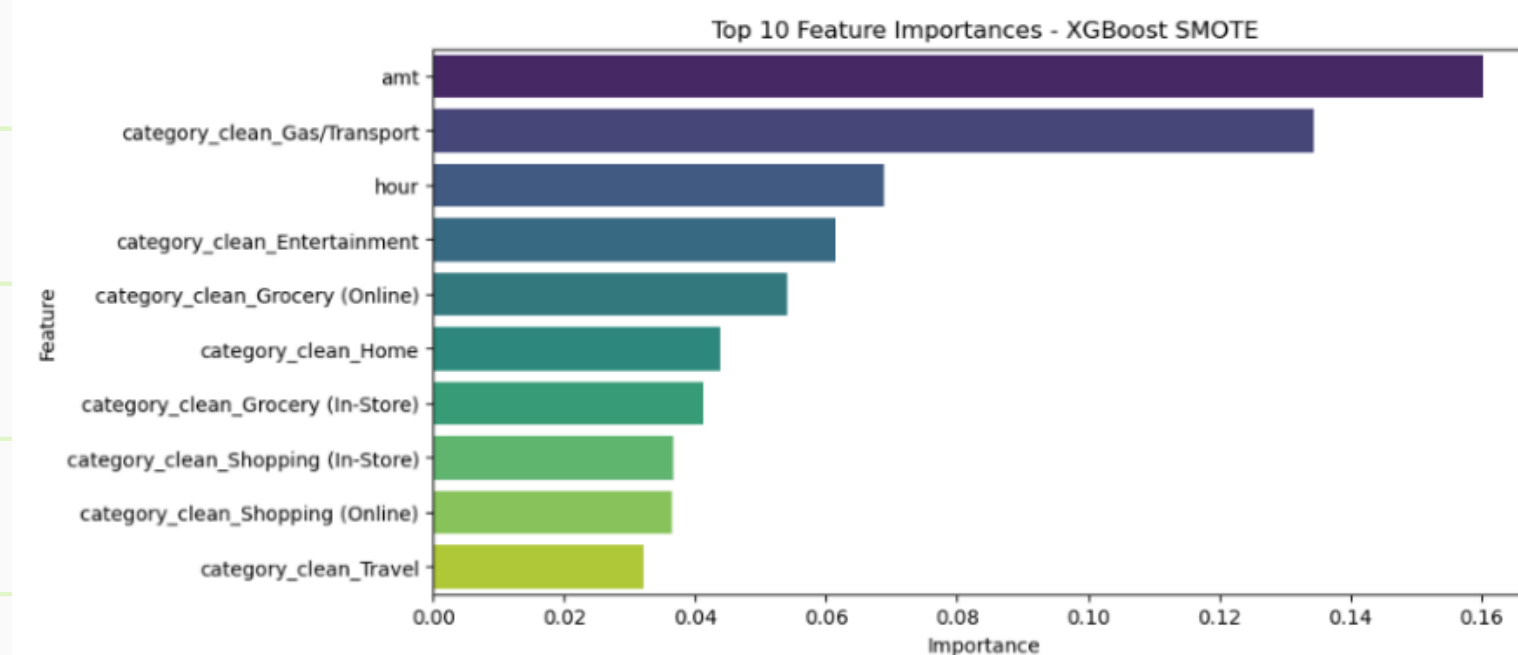
XGBOOST MODEL - SMOTE

```
XGBoost SMOTE Confusion Matrix:  
[[256396  1419]  
 [   101  1419]]  
  
XGBoost SMOTE Classification Report:  
              precision    recall  f1-score   support  
  
     0           1.00       0.99       1.00     257815  
     1           0.50       0.93       0.65       1520  
  
 accuracy              0.99     259335  
 macro avg           0.75       0.96       0.82     259335  
 weighted avg        1.00       0.99       1.00     259335
```

- Recall (fraud): 0.93 → very high detection
- Precision (fraud): 0.50 → best so far
- Strongest balance of precision & recall
- Best overall model performance



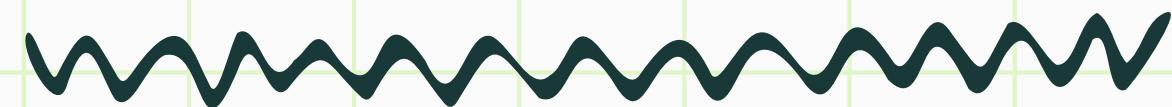
XGBOOST MODEL - SMOTE



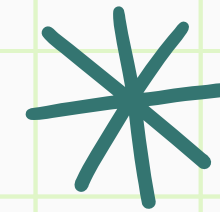
- Transaction amount (amt) is most important
- Category features are highly influential
- Time features (hour) also matter
- Behavioral features > demographics (age, gender)



CONCLUSIONS



- XGBoost + SMOTE is the best model: F1 0.65, recall 0.93, only 101 false positives
- Tree-based models dramatically outperform Logistic Regression for this task
- SMOTE outperforms undersampling, retaining majority class data matters
- Key fraud predictors: transaction amount, hour of day, merchant category
- Cost-sensitive threshold tuning is essential, default 0.5 threshold is wrong for imbalanced data
- Future work:
 - Transaction history features, LSTM sequence models, Concept drift monitoring





THANK YOU

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